A Cross-Sectional Alpha Engine for a 30-Stock Equity Universe

Portfolio Construction, Walk-Forward Evaluation, and Empirical Results

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Abstract

This paper formalizes and evaluates six systematic equity portfolios on a fixed universe of 30 stocks: an equal-weight benchmark, a 12-to-1 momentum portfolio, a defensive machine-learning (ML) portfolio, an ML–momentum blend, a concentrated ML long portfolio, and a dollar-neutral ML long–short portfolio. The strategies are reconstituted from cross-sectional ranks and use either equal or inverse-volatility weights. The supplied walk-forward diagnostics cover January 2019 through May 29, 2026; the 2026 observations are therefore year-to-date rather than a complete calendar year. Momentum Long produced the strongest reported outcome, with a 38.7% compound annual growth rate (CAGR), a 1.52 Sharpe ratio, and a −14.1% maximum drawdown. The Equal Weight Universe was a demanding benchmark, returning 25.7% per year at 16.7% volatility and a 1.46 Sharpe ratio. ML Alpha Long earned 29.7% per year but experienced a −43.4% maximum drawdown and a beta of 1.26. The nominally dollar-neutral ML Alpha L/S portfolio earned −0.1% per year, with a 0.07 Sharpe ratio and a nonzero beta of 0.25. The evidence favors simple momentum in this sample and shows that dollar neutrality, inverse-volatility weighting, and an ML label do not by themselves guarantee factor neutrality, low drawdown, or robust out-of-sample alpha.

Keywords: machine learning, momentum, cross-sectional ranking, inverse-volatility weighting, long–short equity, walk-forward backtest, portfolio turnover.

1 Introduction

The central research question is whether a cross-sectional ML forecast adds economically useful information to simpler portfolio rules in a concentrated equity universe. This is tested through six portfolios that isolate distinct decisions: universe exposure, price momentum, predicted return, defensive tilts, and long–short implementation. All strategies use the same 30-stock opportunity set, so performance differences arise from ranking, selection, weighting, and trading rather than from access to different securities.

The design deliberately includes a strong naive comparator. Equal Weight Universe holds all 30 names at 3.33% each. Although an ML score may sort the securities, the ranking has no portfolio effect because no security is excluded. This portfolio therefore measures the return of the chosen universe without active security selection. It is a more relevant benchmark than cash for judging whether the active rules add value.

The empirical conclusions are sharply differentiated. Pure momentum dominates the reported risk-adjusted statistics, while the ML long–short spread fails to monetize the forecast. The long-only ML portfolios earn high absolute returns but retain substantial market exposure and concentration risk.

The results also show that a strategy called "defensive" can still be riskier than a broad equal-weight benchmark when its defensive signal is applied within a seven-stock portfolio.

This paper makes three contributions. First, it translates the verbal strategy definitions into auditable mathematical rules. Second, it separates selection, weighting, exposure, and cost effects. Third, it interprets the reported results without treating backtest performance as proof of future profitability.

2 Data, Timing, and Research Design

2.1 Universe and rebalance clock

Let $\mathcal{U}_t$ denote the investable universe at rebalance date t, with

$$N_t = |\mathcal{U}_t| = 30. \tag{1}$$

The diagnostics are monthly and run from January 2019 through May 29, 2026. The last reported holdings are dated May 29, 2026. Consequently, any chart labeled 2026 contains only the first five months of that year and must not be compared with complete prior calendar years without adjustment.

Prices should be split- and dividend-adjusted when signals and total returns are calculated. At each month-end, the information set $\mathcal{F}_t$ must contain only data that were observable by that date. A valid walk-forward implementation trains the model on past observations, predicts the next holding-period return, fixes portfolio weights, and then observes the subsequent realized return.

2.2 Prediction target and walk-forward ML score

Let $\mathbf{x}_{i,t}$ be the feature vector for stock i at time t, and let the next-period target be

$$y_{i,t+1} = \frac{P_{i,t+1}}{P_{i,t}} - 1. \tag{2}$$

The generic walk-forward estimator is

$$\widehat{\boldsymbol{\theta}}_t = \arg\min_{\boldsymbol{\theta}} \left\{ \sum_{(i,s)\in\mathcal{T}_t} \mathcal{L}(y_{i,s+1}, f_{\boldsymbol{\theta}}(\mathbf{x}_{i,s})) + \lambda\Omega(\boldsymbol{\theta}) \right\}, \tag{3}$$

where $\mathcal{T}_t$ includes only training observations whose outcomes are known by t, $\mathcal{L}$ is the model loss, and Ω is an optional regularizer. The cross-sectional ML alpha score is the predicted return

$$\widehat{\mu}_{i,t} = f_{\widehat{\boldsymbol{\theta}}_t}(\mathbf{x}_{i,t}). \tag{4}$$

This notation is intentionally model-agnostic: the supplied diagnostics do not identify the estimator, hyperparameters, feature transformations, training-window length, or validation protocol. Those elements must be fixed before a fully reproducible implementation is possible.

2.3 Momentum, volatility, and liquidity signals

The momentum signal excludes the most recent trading month:

$$M_{i,t}^{12\text{-}1} = \frac{P_{i,t-21}}{P_{i,t-253}} - 1. \tag{5}$$

The endpoints approximate the cumulative return from 12 months ago through one month ago. Omitting days $t-20$ through t reduces exposure to short-horizon reversal and prevents the newest price move from dominating the ranking. This construction follows the economic logic of medium-term momentum documented in the asset-pricing literature [1, 2].

For risk control, define the annualized 63-trading-day volatility

$$r_{i,d} = \frac{P_{i,d}}{P_{i,d-1}} - 1, \tag{6}$$

$$\overline{r}_{i,t}^{(63)} = \frac{1}{63}\sum_{j=1}^{63} r_{i,t-j+1}, \tag{7}$$

$$\sigma_{i,t}^{(63)} = \sqrt{252}\left[\frac{1}{62}\sum_{j=1}^{63}\left(r_{i,t-j+1} - \overline{r}_{i,t}^{(63)}\right)^2\right]^{1/2}. \tag{8}$$

Liquidity may be represented by trailing average dollar volume,

$$L_{i,t} = \log\left(\frac{1}{D}\sum_{d=0}^{D-1} P_{i,t-d}V_{i,t-d}\right), \tag{9}$$

where V is share volume and D is the liquidity lookback. The exact D is not stated in the supplied material and is therefore an implementation parameter rather than a verified property of the reported backtest.

2.4 Cross-sectional standardization

Signals with different units must be placed on a common scale before they are combined. For any signal $x_{i,t}$, define its cross-sectional z-score by

$$\mathrm{z}_t(x_{i,t}) = \frac{x_{i,t} - \overline{x}_t}{s_t(x)}, \quad \overline{x}_t = \frac{1}{N_t}\sum_{j\in\mathcal{U}_t} x_{j,t}. \tag{10}$$

The low-volatility score is $-\mathrm{z}_t(\sigma_{i,t}^{(63)})$, so a less volatile stock receives a larger value. In a production process, signal winsorization and a deterministic missing-data policy should precede standardization; otherwise one outlier can materially alter all 30 z-scores.

3 Portfolio Construction

3.1 Selection sets

Let $S_{i,t}$ be a strategy-specific score. For a selection fraction q, define

$$k_q = \lfloor qN_t \rfloor, \qquad \mathcal{H}_t(S,q) = \left\{i : \mathrm{rank}_t^{\downarrow}(S_{i,t}) \le k_q\right\}. \tag{11}$$

With 30 stocks, $k_{20\%} = 6$ and the stipulated $k_{25\%} = 7$. The latter uses the integer floor, consistent with the reported seven holdings. For the short basket,

$$\mathcal{L}_t(S, 20\%) = \left\{i : \mathrm{rank}_t^{\uparrow}(S_{i,t}) \le 6\right\}. \tag{12}$$

Ties should be resolved deterministically, for example by ticker symbol, so repeated runs return identical holdings.

3.2 Equal and inverse-volatility weights

An equal-weight long basket $\mathcal{A}_t$ uses

$$w_{i,t}^{\mathrm{EW}} = \frac{\mathbf{1}\{i \in \mathcal{A}_t\}}{|\mathcal{A}_t|}. \tag{13}$$

An inverse-volatility basket uses

$$w_{i,t}^{\mathrm{IV}} = \frac{\mathbf{1}\{i \in \mathcal{A}_t\}/\sigma_{i,t}^{(63)}}{\sum_{j\in\mathcal{A}_t} 1/\sigma_{j,t}^{(63)}}. \tag{14}$$

Equation (14) reduces the capital allocated to the highest-volatility selected names. It does not target a fixed portfolio volatility and does not account for correlations, so it is not equivalent to minimum-variance optimization.

3.3 The six strategies

Table 1: Strategy definitions. All portfolios have 100% gross exposure.

Strategy	Selection rule	Weighting	Economic role
Equal Weight Universe	All 30 stocks; any ML sorting is economically irrelevant	$1/30 = 3.33\%$ each	Passive universe benchmark
Momentum Long	Top 20% by $M_{i,t}^{12\text{-}1}$	Six stocks at $1/6 = 16.67\%$ each	Pure medium-term price trend
Defensive Alpha	Top 25% by $0.50z(\widehat{\mu}) - 0.30z(\sigma) + 0.20z(L)$	Seven inverse-63-day-volatility weights	ML forecast with low-risk and liquidity preferences
Alpha + Momentum	Top 25% by $0.60z(\widehat{\mu}) + 0.40z(M^{12\text{-}1})$	Seven inverse-63-day-volatility weights	Explicitly gives momentum additional weight beyond its presence among the ML features
ML Alpha Long	Top 20% by $\widehat{\mu}$	Six stocks at 16.67% each	Concentrated long-only ML portfolio
ML Alpha L/S	Top 20% long and bottom 20% short by $\widehat{\mu}$	Six at $+1/12$ and six at $-1/12$	50% long, 50% short; dollar-neutral spread

The two composite scores are therefore

$$S_{i,t}^{\mathrm{Def}} = 0.50\,\mathrm{z}_t(\widehat{\mu}_{i,t}) - 0.30\,\mathrm{z}_t(\sigma_{i,t}^{(63)}) + 0.20\,\mathrm{z}_t(L_{i,t}), \tag{15}$$

$$S_{i,t}^{\mathrm{AM}} = 0.60\,\mathrm{z}_t(\widehat{\mu}_{i,t}) + 0.40\,\mathrm{z}_t(M_{i,t}^{12\text{-}1}). \tag{16}$$

Volatility enters Defensive Alpha twice: once through the $-0.30z(\sigma)$ selection term and again through the inverse-volatility weights. Momentum likewise enters Alpha + Momentum both indirectly, because it is among the ML features, and directly through the 40% composite-score weight.

For ML Alpha L/S, the weights are

$$w_{i,t}^{\mathrm{LS}} = \frac{1}{12}\mathbf{1}\{i \in \mathcal{H}_t(\widehat{\mu}, 20\%)\} - \frac{1}{12}\mathbf{1}\{i \in \mathcal{L}_t(\widehat{\mu}, 20\%)\}. \tag{17}$$

Thus $\sum_i |w_{i,t}| = 1$ and $\sum_i w_{i,t} = 0$. These identities establish gross and dollar exposure only. If the long and short baskets have unequal market betas, sector weights, duration, or style exposures, the portfolio is not factor-neutral.

3.4 Returns, trading, and costs

For weights fixed at time t, the next-period gross portfolio return is

$$R_{p,t+1}^{\text{gross}} = \sum_{i \in \mathcal{U}_t} w_{i,t} r_{i,t+1}. \tag{18}$$

Let $\widetilde{w}_{i,t}$ be the weight immediately before rebalancing after market drift. One-way turnover is

$$\text{TO}_t = \frac{1}{2} \sum_i |w_{i,t} - \widetilde{w}_{i,t}|. \tag{19}$$

A general net-return specification is

$$R_{p,t+1}^{\text{net}} = R_{p,t+1}^{\text{gross}} - c_t \text{TO}_t - \sum_{i: w_{i,t} < 0} b_{i,t} |w_{i,t}|, \tag{20}$$

where c_t represents commissions, spread, slippage, and market impact per unit of turnover, while $b_{i,t}$ is the short-borrow cost. The supplied plots label performance as net of costs but do not disclose c_t, $b_{i,t}$, or capacity assumptions. The reported net results must therefore be understood as conditional on an undocumented cost model.

4 Performance Measurement

Let R_t denote monthly net portfolio returns and let T be the number of months. Wealth evolves as

$$W_T = W_0 \prod_{t=1}^{T} (1 + R_t). \tag{21}$$

The reported statistics can be defined as follows:

$$\text{CAGR} = \left(\frac{W_T}{W_0} \right)^{12/T} - 1, \tag{22}$$

$$\sigma_{\text{ann}} = \sqrt{12}\, \text{sd}(R_t), \tag{23}$$

$$\text{Sharpe} = \sqrt{12}\, \frac{\overline{R_t - R_{f,t}}}{\text{sd}(R_t - R_{f,t})}, \tag{24}$$

$$D_t = \frac{W_t}{\max_{s \le t} W_s} - 1, \qquad \text{MaxDD} = \min_t D_t, \tag{25}$$

$$\text{CVaR}_{95\%} = -\mathbb{E}[R_t \mid R_t \le q_{0.05}(R)]\,, \tag{26}$$

$$N_{\text{eff},t} = \frac{1}{\sum_i w_{i,t}^2}, \tag{27}$$

$$\beta_p = \frac{\text{Cov}(R_{p,t}, R_{\text{EW},t})}{\text{Var}(R_{\text{EW},t})}. \tag{28}$$

CVaR is displayed as a positive loss magnitude. Effective N measures concentration: it equals 30 for a 30-name equal-weight portfolio, 6 for a six-name equal-weight portfolio, and 12 for the twelve positions in the $\pm 8.33\%$ long–short spread. Beta is measured relative to the Equal Weight Universe, not to an external market index.

5 Empirical Results

5.1 Headline scorecard

Table 2 reproduces the exact labels in the supplied portfolio scorecard. The sample ends on May 29, 2026.

Table 2: Reported walk-forward net performance, January 2019–May 29, 2026.

Strategy	CAGR	Volatility	Sharpe	Max DD	CVaR 95%	Turnover	Effective N	Beta
ML Alpha L/S	−0.1%	14.5%	0.07	−33.5%	8.7%	92.7%	12.00	0.25
ML Alpha Long	29.7%	26.4%	1.12	−43.4%	13.8%	83.5%	6.00	1.26
Alpha + Momentum	31.0%	22.7%	1.31	−23.6%	10.5%	79.0%	6.68	1.07
Defensive Alpha	26.6%	22.2%	1.18	−26.4%	9.8%	98.1%	6.58	1.07
Momentum Long	38.7%	23.6%	1.52	−14.1%	8.6%	43.1%	6.00	1.03
Equal Weight Universe	25.7%	16.7%	1.46	−18.2%	8.0%	1.1%	30.00	1.00

Momentum Long is the strongest portfolio on the principal performance measures. It exceeds the equal-weight benchmark by 13.0 percentage points of CAGR, has the highest Sharpe ratio, and has the shallowest maximum drawdown. Its 43.1% turnover is also materially below every active ML strategy. This is a demanding result for the alpha engine because momentum is both simpler and less trading-intensive.

Alpha + Momentum is the best ML-containing long-only strategy by Sharpe ratio and earns 5.3 percentage points more CAGR than Equal Weight. However, it does not beat pure momentum. This suggests that, over the reported sample, the ML component either diluted the explicit trend signal or added positions whose incremental return did not compensate for additional turnover and drawdown.

Defensive Alpha earns only 0.9 percentage points more CAGR than Equal Weight while taking more volatility, a deeper drawdown, greater CVaR, and 98.1% turnover. Its label is best interpreted relative to ML Alpha Long, not relative to the broad benchmark. Against ML Alpha Long, the defensive construction lowers volatility by 4.2 percentage points, improves maximum drawdown by 17.0 percentage points, and reduces CVaR by 4.0 percentage points.

ML Alpha Long generates a high 29.7% CAGR but has the weakest downside profile among the long-only portfolios: 26.4% volatility, −43.4% maximum drawdown, 13.8% CVaR, and beta of 1.26. The result is consistent with a concentrated, high-market-exposure portfolio rather than a low-risk source of standalone alpha.

ML Alpha L/S is the clearest negative finding. Despite zero net dollar exposure, it delivers a −0.1% CAGR and a 0.07 Sharpe ratio while experiencing a −33.5% maximum drawdown. Its beta of 0.25 confirms that dollar neutrality did not eliminate systematic exposure. Possible explanations include weak ordering among the bottom-ranked names, unstable cross-sectional forecasts, asymmetry between long and short signal efficacy, short costs, and uncompensated factor tilts.

5.2 Wealth and drawdown paths

Figure 1 shows the growth of one dollar net of the backtest's assumed costs. Momentum separates from the other portfolios after 2020 and finishes at more than eleven times initial capital. The long-only

alternatives finish between roughly five and seven times initial capital, whereas ML Alpha L/S remains close to its starting value.

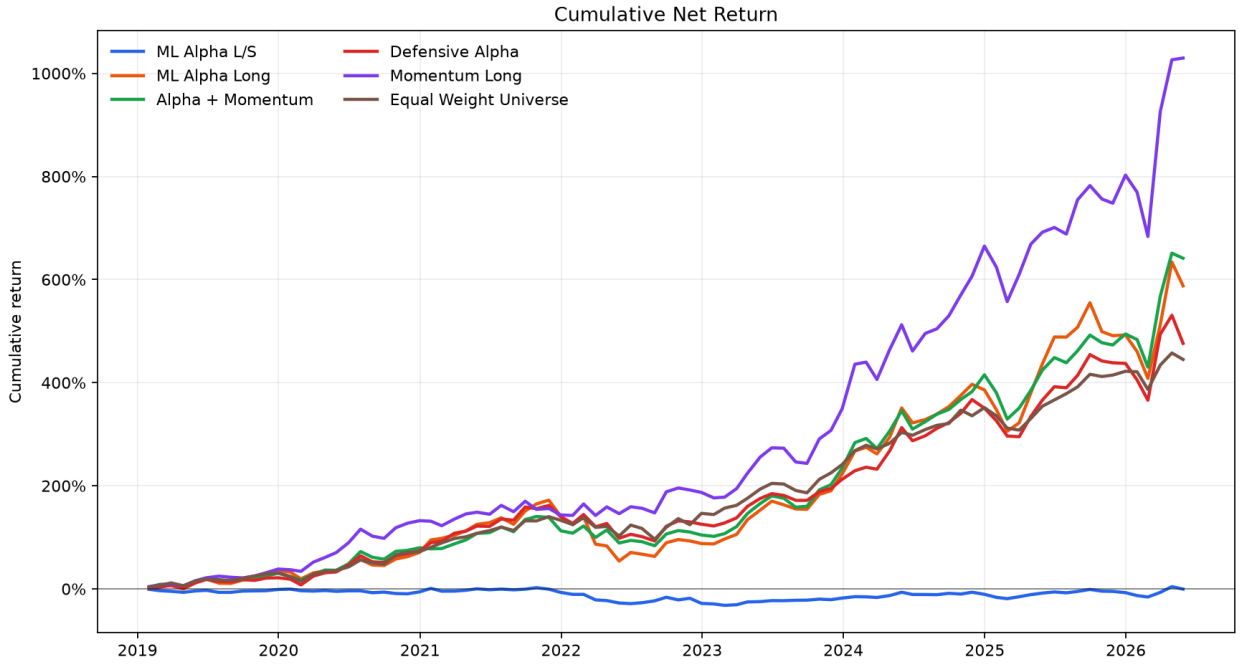


Figure 1: Cumulative net return. Source: supplied Alpha Engine diagnostic pack.

The path dependence is economically important. Figure 2 shows that the 2022 stress period produced the deepest losses for ML Alpha Long and ML Alpha L/S. Momentum's maximum drawdown remained considerably smaller, and the equal-weight benchmark also proved more resilient than most ML portfolios. A high terminal CAGR did not imply a smooth or easily financeable investment path.

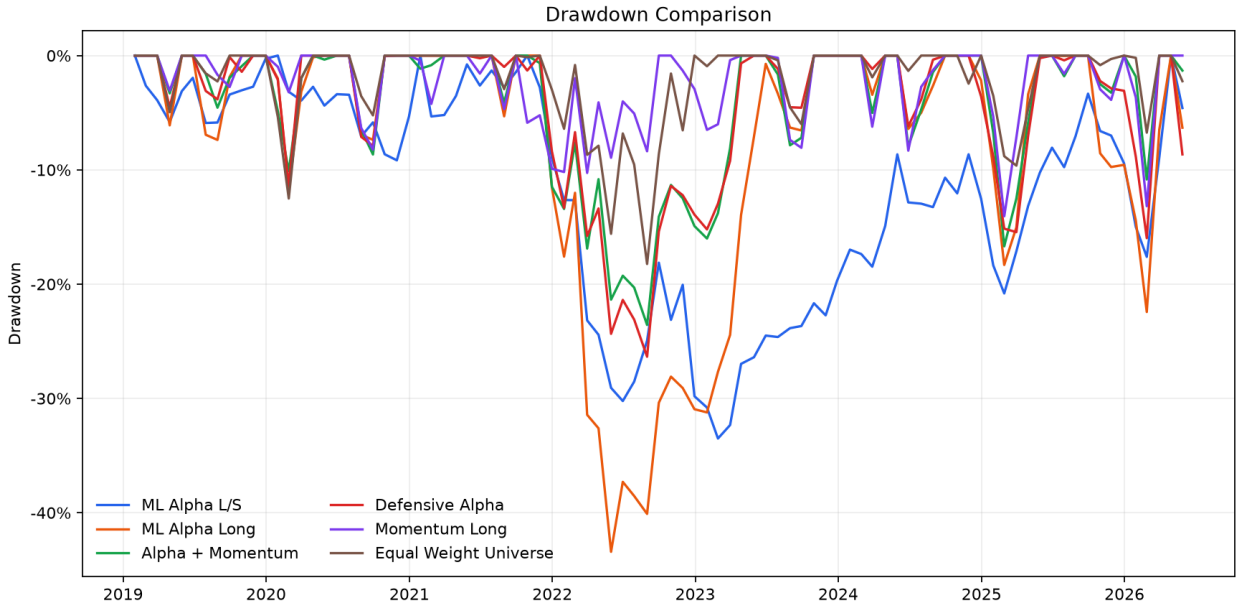


Figure 2: Drawdown comparison. Source: supplied Alpha Engine diagnostic pack.

The calendar-year decomposition in Figure 3 further shows regime dependence. The figure's 2026 bars are partial-year returns through May 29, 2026. In 2022 the ML long and long–short portfolios suffered large losses, while Momentum Long and Equal Weight remained positive. Strong rebounds in 2023 and 2024 account for a substantial part of the long-only portfolios' terminal wealth.

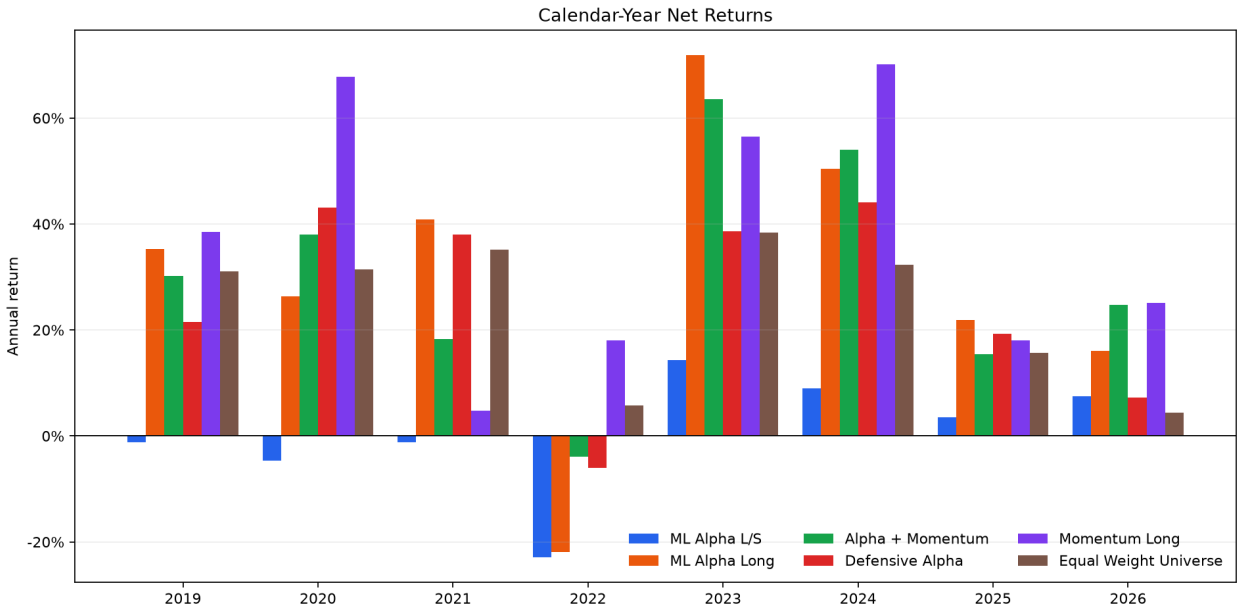


Figure 3: Calendar-year net returns. The 2026 bar is year-to-date through May 29. Source: supplied Alpha Engine diagnostic pack.

5.3 Dependence and market exposure

Monthly-return correlations reveal that most long-only portfolios are variations on a common equity exposure. Alpha + Momentum has correlations of 0.93 with Momentum Long, 0.88 with Defensive Alpha, 0.87 with ML Alpha Long, and 0.79 with Equal Weight. By contrast, ML Alpha L/S has only 0.28 correlation with Equal Weight. Its diversification benefit is real in a correlation sense, but its near-zero return and substantial drawdown mean that low correlation alone is insufficient for inclusion.

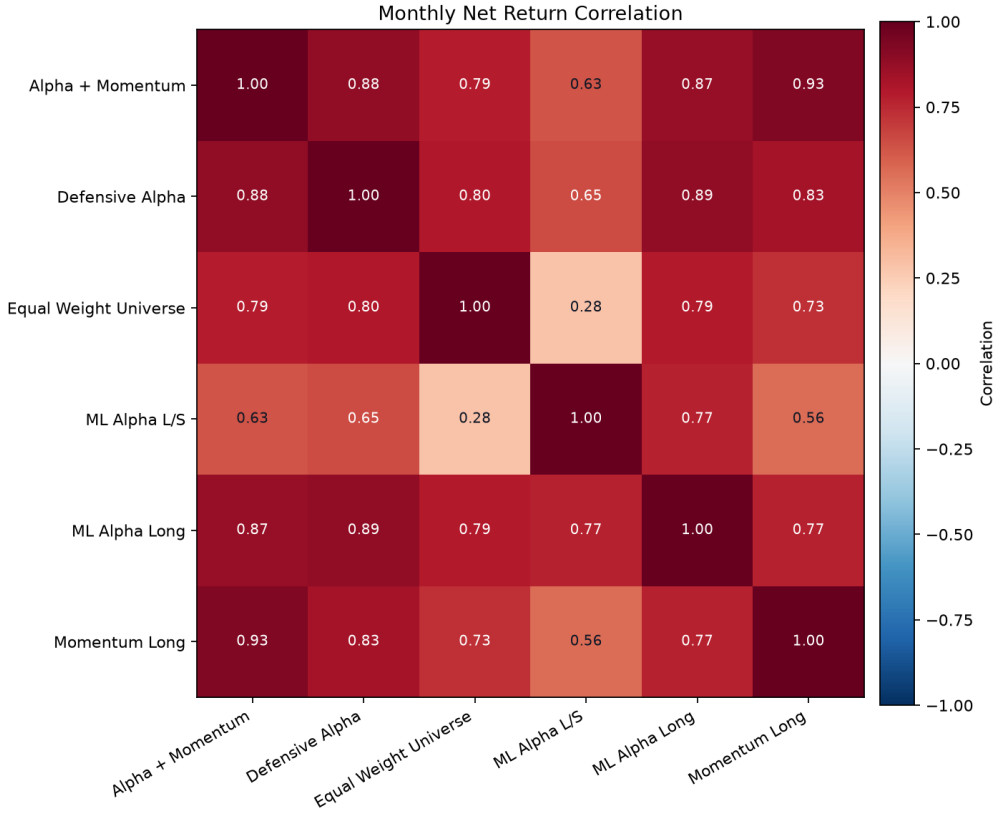


Figure 4: Correlation of monthly net returns. Source: supplied Alpha Engine diagnostic pack.

Full-sample betas conceal instability. Figure 5 indicates that rolling 12-month beta changed materially over time, particularly around 2022–2024 and again near the end of the sample. Late-sample betas rose sharply for several long-only strategies. Risk budgets based only on the full-sample betas in Table 2 would therefore understate time variation in systematic exposure.

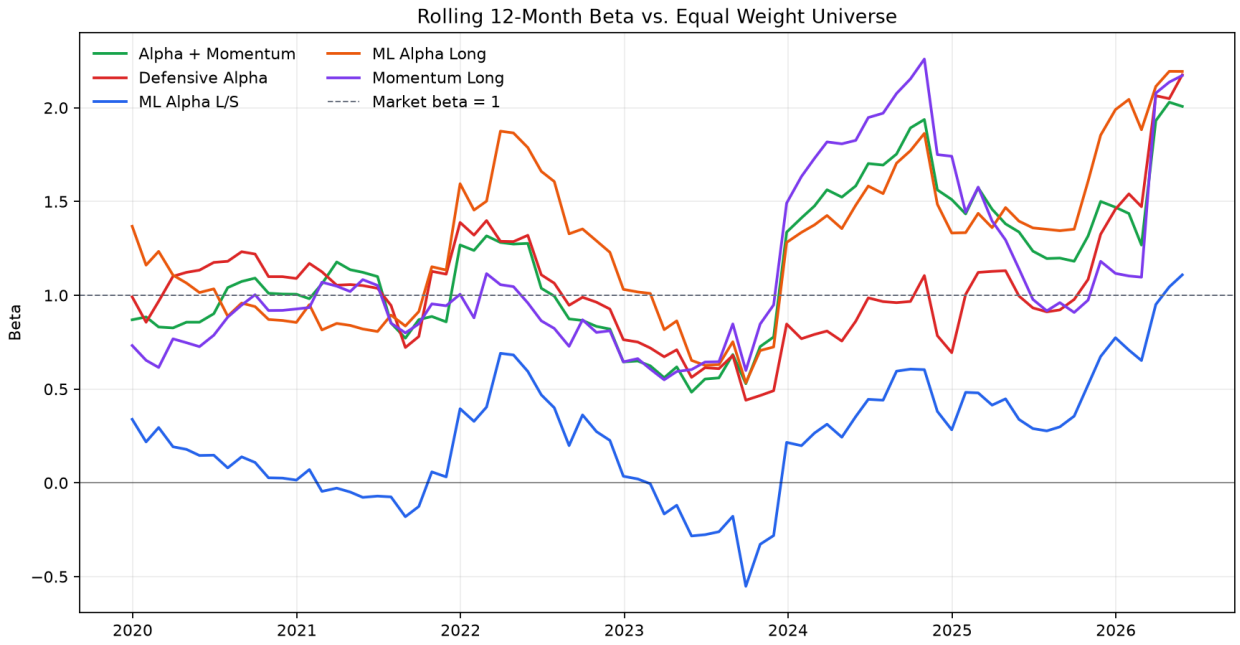


Figure 5: Rolling 12-month beta relative to Equal Weight Universe. Source: supplied Alpha Engine diagnostic pack.

5.4 Last observed portfolio holdings

Table 3 transcribes the holdings at the last reported rebalance, May 29, 2026. It is a historical snapshot, not a recommendation or a current portfolio as of August 6, 2026.

Table 3: Last observed active holdings on May 29, 2026.

Strategy	Holdings and target weights
ML Alpha L/S	Long: LLY, GE, AMD, CRM, ADBE, ORCL at +8.33% each. Short: BAC, AAPL, JPM, PEP, KO, V at −8.33% each.
ML Alpha Long	LLY, GE, AMD, CRM, ADBE, ORCL at 16.67% each.
Alpha + Momentum	GOOGL 17.8%, NVDA 17.4%, CAT 15.0%, AVGO 14.8%, GE 14.7%, ORCL 10.7%, AMD 9.5%.
Defensive Alpha	MSFT 19.2%, GOOGL 16.9%, NVDA 16.3%, LLY 15.5%, CRM 13.2%, ORCL 10.0%, AMD 8.9%.
Momentum Long	XOM, NVDA, AVGO, GOOGL, CAT, AMD at 16.67% each.

The ML Alpha Long basket is identical to the long side of ML Alpha L/S by construction. This makes the long–short failure especially informative: the difference between those strategies is the short basket and the reduction of long exposure from 100% to 50%. The combined spread can fail even when the long ranking is useful if the predicted losers rise, the short side has adverse factor exposure, or the reduced long exposure sacrifices the equity premium.

6 Economic Interpretation

6.1 Why momentum may have dominated

The 12-to-1 momentum rule has several structural advantages in this experiment. It is transparent, stable, and based on a signal whose ordering can persist for multiple months. That persistence is consistent with its lower turnover. The one-month skip also reduces contamination by very recent reversals. In a 30-stock universe, a complex model has limited cross-sectional breadth; a six-stock top basket can change materially when only one or two forecast ranks move. A simple trend measure may therefore have a better signal-to-estimation-error ratio.

The finding does not prove that ML contains no information. Alpha + Momentum and ML Alpha Long both exceed the equal-weight benchmark in CAGR. Rather, the evidence indicates that the incremental ML forecast was not superior to momentum after accounting for portfolio path and trading intensity in this particular sample.

6.2 Why defensive construction did not beat the benchmark on risk

Defensive Alpha applies low volatility twice, but only after concentrating the universe into seven stocks. Selection concentration can dominate within-basket risk scaling. Inverse-volatility weighting also ignores correlations: seven individually calm stocks can share a common industry or factor exposure. Finally, the ML and liquidity components can select securities that are low-volatility only relative to the active subset, not relative to the 30-stock benchmark. These mechanisms explain why Defensive Alpha is safer than ML Alpha Long but riskier than Equal Weight.

6.3 Why dollar neutrality did not produce alpha neutrality

Equation (17) guarantees $\sum_i w_i = 0$, not $\sum_i w_i \beta_i = 0$. Portfolio beta is approximately

$$\beta_{p,t} \approx \sum_i w_{i,t} \beta_{i,t}, \tag{29}$$

which is nonzero whenever the beta-weighted long exposure differs from the beta-weighted short exposure. The same logic applies to sectors and style factors. A factor-neutral extension would solve for weights subject to constraints such as

$$\mathbf{1}'\mathbf{w} = 0, \qquad \boldsymbol{\beta}'\mathbf{w} = 0, \qquad \mathbf{B}'\mathbf{w} = \mathbf{0}, \tag{30}$$

where $\mathbf{B}$ contains sector and style loadings. Such constraints may reduce unintended risk, although they can also weaken the raw alpha signal and increase turnover.

7 Robustness, Limitations, and Required Validation

The reported findings are economically interesting but not sufficient for a production decision. The following tests are required before capital allocation:

- **Reproducibility.** The supplied Word file contains chart images rather than raw return observations, and the companion alpha-engine artifacts contain no code or data. The scorecard can be transcribed but not independently recomputed from the supplied package.

- **Universe integrity.** A historically fixed 30-stock list can create survivorship and look-ahead bias if membership was selected using information from the end of the sample. Point-in-time membership and delisted securities should be retained.
- **ML protocol.** The estimator, features, target horizon, training window, hyperparameter search, embargo, and model-refit frequency must be documented. Every transformation must be fitted using training data only.
- **Costs and capacity.** Results should be rerun over a grid of spreads, slippage, borrow fees, and assets under management. The 79–98% reported turnover of the active ML portfolios makes cost sensitivity central rather than cosmetic.
- **Statistical uncertainty.** January 2019 through May 2026 provides only about 89 monthly observations and includes an unusually strong large-cap equity period. Block-bootstrap confidence intervals and multiple-testing corrections are needed; overlapping strategy variants should not be treated as independent discoveries [3].
- **Benchmarking.** Returns should be regressed on an external broad-market factor and standard equity factors in addition to the internal equal-weight benchmark. Dollar-neutral and residualized versions should be evaluated separately.
- **Signal ablation.** Compare ML without momentum features, momentum alone, ML plus momentum, and defensive terms one at a time. This identifies whether the ML model contributes information beyond the explicitly reintroduced momentum signal.
- **Regime and stability analysis.** Repeat the test across pre-2019 data, alternative universes, rebalance frequencies, volatility windows, and selection fractions. Report rank information coefficients, top-minus-bottom spreads, and sector exposures through time.

A particularly important ablation is the long–short decomposition:

$$R_t^{\mathrm{LS}} = \frac{1}{2} R_t^{\mathrm{LongBasket}} - \frac{1}{2} R_t^{\mathrm{ShortBasket}}. \tag{31}$$

Reporting each leg separately would determine whether the problem lies in weak long forecasts, rising predicted losers, short implementation costs, or the loss of market exposure.

8 Conclusion

The six-portfolio experiment separates the value of the universe, the signals, and the implementation rules. Over January 2019 through May 29, 2026, the 30-stock equal-weight universe was already a strong investment, compounding at 25.7% with a 1.46 Sharpe ratio. Pure 12-to-1 momentum added the most value: 38.7% CAGR, 1.52 Sharpe, and the shallowest drawdown. Alpha + Momentum improved on the benchmark but not on momentum alone. Defensive Alpha reduced risk relative to the concentrated ML long portfolio but remained riskier and more expensive to trade than Equal Weight. ML Alpha Long generated high absolute return largely alongside substantial market and concentration exposure. ML Alpha L/S did not deliver a viable standalone alpha stream despite its dollar-neutral construction.

The practical conclusion is not that ML should be discarded. It is that the ML forecast must demonstrate incremental information beyond simple momentum, survive a fully specified cost model, and be neutralized against unintended risks when used in a long–short portfolio. Until raw data and

code reproduce the reported figures and the required robustness tests are passed, Momentum Long and Equal Weight should be treated as the principal reference strategies, while the ML portfolios remain research candidates rather than validated production allocations.

A Walk-Forward Implementation Checklist

At every rebalance date t, a reproducible engine should execute the following sequence:

1. Freeze the point-in-time universe and all data available by t.
2. Apply corporate-action adjustments and enforce publication lags for non-price features.
3. Fit preprocessing and the ML model using only the permitted training window.
4. Generate $\widehat{\mu}_{i,t}$ and compute momentum, volatility, and liquidity.
5. Winsorize using a predeclared rule, then compute cross-sectional z-scores.
6. Rank names, apply the exact integer portfolio sizes, and resolve ties deterministically.
7. Form equal or inverse-volatility weights and verify gross, net, name, and sector limits.
8. Trade from drifted weights, record turnover, costs, borrow availability, and realized fills.
9. Hold weights fixed until the next rebalance and store the realized net return.
10. Save predictions, ranks, weights, exposures, and model metadata for an audit trail.

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